Code-Switching Translation with PEFT on a Frozen Qwen3-1.7B

Midterm Project · Advanced Topics in Large Language Models · Student ID 2026711223

1. Overview

We build two parameter-efficient translation systems on a frozen **Qwen3-1.7B** backbone that translate single-language or code-switched English/Spanish/Korean/Japanese input into English or Korean. Both systems are LoRA adapters on all linear projections of all 28 layers: the **5M system** uses rank 4 on every projection (4,358,144 trainable parameters) and the **50M system** an MLP-heavy allocation, rank 20 on the attention projections and rank 60 on the MLP projections (49,315,840). Because no code-switched translation corpus exists for these languages, we synthesise the training data reference-first: an LLM writes clean sentences with faithful four-language versions and two code-switched variants whose matrix language, embedded language(s) and switching pattern we control explicitly (noun insertion with native particles, clause alternation, phrase mixing, light-verb constructions); variants are verified by an English-pivot round trip with an independent local model. On the official validation set (500 items) the untuned baseline scores 52.26; the 5M system reaches 67.30 and the 50M system 68.46 (+15.0 / +16.2 chrF++), with the 50M system ahead on three of the four subsets there and on every subset of our two own development sets.

2. PEFT design

Both systems add LoRA adapters (Hu et al., 2021) to **every linear projection of every decoder layer** of the frozen Qwen3-1.7B backbone (28 layers; q/k/v/o projections of the attention block and gate/up/down projections of the MLP), with `lora_alpha = 2r`, no dropout, no bias, and the backbone weights never updated. Qwen3-1.7B has hidden size 2048, MLP size 6144 and 8 KV heads of dimension 128, so one rank of LoRA on all seven projections of one layer costs 4096+3072+3072+4096+8192+8192+8192 = 38,912 parameters, i.e. 1,089,536 parameters per unit of rank for the whole network.

System	Rank r	alpha	Modules	Trainable params	Budget use
5M	4	8	q,k,v,o,gate,up,down × 28 layers	4,358,144	87.2%
50M	20 (attn) / 60 (MLP)	40 / 120	q,k,v,o × 28 layers at r=20; gate,up,down × 28 layers at r=60	49,315,840	98.6%

Motivation. Under a fixed parameter budget the literature consistently favours spreading low-rank adapters over all linear layers (in particular the MLP) rather than concentrating a high rank on attention only (LoRA module ablations; QLoRA's "all linear layers" finding; Biderman et al., 2024). r=4 is the largest uniform all-linear rank that fits 5M. For 50M we first used the largest uniform rank, r=44 (47,939,584 parameters), and then re-allocated the budget towards the MLP (attention r=20, MLP r=60, implemented with PEFT's `rank_pattern`/`alpha_pattern`): the MLP projections are the largest matrices and hold most of the backbone's lexical knowledge, which is where Korean generation needs capacity, and on the official validation set the MLP-heavy adapter beat the uniform one by +0.9 at equal data (Section 5.4). The 50M system is thus a strict superset of the 5M system (same modules, same data, same recipe) with 5× the attention rank and 15× the MLP rank. We also tried DoRA at r=4 (4,931,584 parameters), an MLP-heavy 5M variant (attention r=2 / MLP r=5, 4,243,456 parameters) and a learning-rate sweep; see Sections 5.4 and 6.

Training. Supervised fine-tuning on (prompt, translation) pairs rendered with the Qwen3 chat template in non-thinking mode, i.e. the prompt ends with `<|im_start|>assistant\n<think>\n\n</think>\n\n` exactly as at inference, and the loss is computed only on the translation tokens plus `<|im_end|>`. AdamW, cosine schedule, 3% warm-up, effective batch 16, bf16, gradient checkpointing, 2 epochs (lr 3e-4 for 5M, 2e-4 for 50M); one RTX 3080 (12 GB) trains ~2,300 tokens/s, i.e. ~5.9 h per final system over the 244,852 training rows.

3. Training data generation

No suitable code-switched translation corpus exists for en/es/ko/ja, so all training data is synthetic and built **reference-first**: a clean target-language sentence is written first and the code-switched input is derived from it, so the reference never contains generation noise and stays literal, which is what a single-reference chrF++ rewards.

Step 1 – scenarios. 48 domains (business meetings, finance, hospital, pharmacy, airports, hotels, restaurants, IT operations, logistics, customs, education, HR, legal, manufacturing, retail, ...) × 30 concrete situations with two participant roles each (1,440 scenarios), generated by an LLM (Claude Sonnet via the headless `claude -p` CLI; the same driver works with a local model through vLLM, `scripts/gen_vllm.py`).

Step 2 – 4-way tuples with controlled switching. For each scenario the LLM writes 8 different sentences (7–30 words, varied speech acts, concrete dates/amounts/names, no idioms), each with a faithful literal English, Korean (polite -습니다/-세요 register), Spanish and Japanese (です/ます) version, plus **two code-switched variants**. Every variant follows a per-sentence specification `matrix=X; embed=Y[,Z]; pattern=P` sampled by us (matrix language ko 35% / en 25% / ja 20% / es 20%; one embedded language 70%, two 30%; pattern *noun* 50%, *clause* 22%, *phrase* 18%, *verb* 10%). The matrix language supplies grammar, word order and all particles; embedded spans are copied

verbatim from the parallel sentence. The patterns mirror the specification's examples: noun/NP insertion with native particles attached ("chest pain이", "meetingが", "El informe의 마지막 section을"), clause alternation at a clause boundary ("Aunque el tren llegó tarde, 우리는 회의에 늦지 않았습니다."), phrase mixing, and Korean/Japanese light-verb constructions ("주문을 cancel하고", "confirmしてください"). This design follows the linguistic evidence that natural ko–en / ja–en code-switching is predominantly insertional (nouns with matrix-language morphology) and that LLM generation is only reliable with explicit constraints. Tuples were generated in four waves that differ in which language is written first and in the switching specification: w1 (3,201 tuples, English-first, model-chosen switch points), v2 (11,521, English-first with the explicit specifications above), v2ko (11,520, Korean-first, so the Korean reference is the author's own sentence rather than a translation) and v3 (10,240, Spanish- or Japanese-first, one to three embedded languages, sentence shapes matched to the official validation inputs): 36,482 raw tuples in total.

Step 3 – filtering. (i) script checks: Korean fields must be Hangul-only, Japanese fields must contain kana/kanji, and the English/Spanish fields must be identified as English/Spanish by a language identifier (this removed ~2% of tuples whose en/es fields were swapped); (ii) each mixed variant must actually contain ≥2 scripts (or en+es by LID); (iii) an **English-pivot round trip**: every mixed variant is back-translated into English by an independent local model (Qwen3-14B-AWQ, vLLM) and variants with chrF++ < 40 against the tuple's English sentence are dropped (≈2.4%); (iv) exact de-duplication and removal of any sentence overlapping the development set or the official validation/test inputs and references.

Step 4 – assembling rows. Each surviving mixed variant yields two rows (→ English, → Korean, references = the tuple's en / ko sentences); the four monolingual versions yield single-language rows (es/ja/ko → en, en/es/ja → ko), sub-sampled so that ~30% of the rows are single-language; ~5% of tuples also contribute an identity row (en→en or ko→ko) so the model learns to copy text already in the target language. The final set has 36,334 clean tuples → 204,852 synthetic rows (mixed→en 71,698; mixed→ko 71,699; single→en 30,794; single→ko 30,661, of which 523 identity rows), plus 40,000 human-translated sentence pairs (AI Hub ko–en daily/general/specialised and ko–ja, BSD business dialogues ja–en, Tanaka ja–en, TED2020 es→ko/es→en/ja→ko/ko→en; see Section 5.2 for why) = 244,852 rows. Mixed rows cover all six language pairs (6.3–9.6k tuples each), all four triples (5.0–6.0k each) and the four-language mix (3.7k); 36% of mixed rows use three or more languages.

Step 5 – reference style. The official validation references (Section 5.5) showed that natural, concise Korean without commas is what the metric rewards, so the Korean references were post-edited: 14,701 Korean reference sentences (those of rows whose input contains no Korean) were rewritten by the LLM into natural polite Korean under a strict keep-every-fact instruction (`prompts/ko_natural.txt`), which replaced the reference of 23,428 →ko rows (12,678 distinct sentences; the same rewrites also serve as the source of 4,120 ko→en rows), and commas were removed from every Korean reference, synthetic and human. Rows whose input already contains Korean keep the literal reference so that Korean spans are copied, not paraphrased. The validation references themselves were never used for training.

Development set. Because the official validation data was not yet available, we built our own: 384 tuples generated with a different prompt and different scenario seeds (and, as a robustness check, a second set generated by a different model family, Qwen3-8B), giving 1,535 items: mixed→en 384, mixed→ko 383, single→en 384, single→ko 384. All model selection used this set.

4. Inference design

The instruction is the default template from the specification (target language inserted), wrapped in the Qwen3 chat template with `enable_thinking=False`; no few-shot examples, no system prompt. Decoding is greedy (`do_sample=False`), at most 256 new tokens, stopping on `<|im_end|>`; the adapter is applied unmerged with HF transformers (we found vLLM's LoRA path changes ~25% of greedy outputs, so all reported numbers and the submitted predictions use the same HF path). Predictions are NFC-normalised and stripped; for Korean targets commas are removed (see Section 5.5).

5. Results

All numbers are corpus-level chrF++ (sacrebleu `CHRF(word_order=2)`) under the official aggregation: single = mean(single→en, single→ko), mixed = mean(mixed→en, mixed→ko), score = 0.3·single + 0.7·mixed. The baseline is the untuned Qwen3-1.7B with the default instruction template, greedy, non-thinking. The main results are on the instructor's validation set (500 items: 150 single→en, 150 single→ko, 100 mixed→en, 100 mixed→ko; single inputs are Spanish, Japanese, Korean or English sentences, mixed inputs cover all language pairs and many three- and four-language mixes). Its references were never used for training; it served only to choose between the configurations compared in 5.4.

5.1 Main results (official validation set)

System	params	single→en	single→ko	mixed→en	mixed→ko	single	mixed	score
Baseline (no PEFT)	0	59.15	28.04	69.17	42.78	43.59	55.98	52.26

5M (LoRA r=4, all linear)	4,358,144	63.68	48.46	74.46	69.78	56.07	72.12	**67.30**
50M (LoRA attn r=20 / MLP r=60)	49,315,840	62.73	50.27	75.04	72.12	56.50	73.58	**68.46**
Copy input	–	6.35	1.15	27.28	31.38	3.75	29.33	21.66

Both systems clear the baseline by a wide margin (+15.0 and +16.2 points; Success criterion 1). The gain is largest on Korean targets (+20 single→ko, +27/+29 mixed→ko) because that is where the untuned model fails outright (Section 6). The 50M system is ahead of the 5M system on single→ko (+1.8), mixed→en (+0.6) and mixed→ko (+2.3) and 1.0 behind on single→en. At the sentence level the two systems tie on that subset (the 50M output scores higher on 63 of the 150 items, lower on 61): the corpus-level gap comes from a handful of Spanish sentences that the 50M system renders more freely ("without having completed the payment" → "before paying", "comes from less polluting sources" → "comes from cleaner sources") while the 5M system stays literal, which a single-reference chrF++ rewards. On both of our own development sets the 50M system is ahead on all four subsets (5.3). Both systems were trained for 2 epochs; the epoch-1 checkpoints scored 67.04 (5M) and 66.64 (50M), and averaging the epoch-1 and epoch-2 adapter weights did not help (67.18 / 67.85).

5.2 How the score grew with data

Own dev set (1,535 items): training tuples (rows)	5M	50M (r=44)
0 (baseline)	53.98	53.98
447 (2.5k rows, pilot)	62.16	63.18
6.3k (35.6k rows, round 1)	70.19	71.70
6.3k + 30k human pairs	70.78	71.41
26.2k + 40k human (187k rows)	73.43	74.54

Official validation set: final rounds	rows	5M (r=4)	50M (attn 20 / MLP 60)
0 (baseline)	–	52.26	52.26
26.2k tuples + 40k human, literal Korean references	187,252	65.81	66.86
same, natural comma-free Korean references (Step 5)	187,252	66.16	66.24
+ v3 wave (36.3k tuples) = final systems	244,852	**67.30**	**68.46**

Even 447 synthetic tuples lift the own-dev score by 8 points, mostly by removing the baseline's Korean-side failures (instruction echoing, repetition loops, answering in the wrong language). Adding 30k human-translated sentence pairs on top of the synthetic data gives a further +0.6 on our dev set and +0.8 on the second dev set, so the final systems include 40k of them. On the official set, rewriting the Korean references into natural style moved the two systems by +0.4 / −0.6 (inside the ±0.7 sampling noise of a 500-item set) but made the 50M system dominate the 5M system on both own dev sets (5.3); adding the v3 wave (10,240 Spanish/Japanese-first tuples whose sentence shapes match the validation inputs, +31% rows) gave +1.1 / +2.2, the largest single step of the final rounds. Data, not adapter configuration, remained the lever throughout.

5.3 Own development sets

Before the official set arrived, all selection used our own dev set (Section 3). Round-1 systems on it: baseline 53.98 (66.08 / 31.85 / 72.74 / 39.50 for single→en / single→ko / mixed→en / mixed→ko), 5M 73.43 (78.79 / 58.19 / 83.79 / 67.31), 50M r=44 74.54 (79.98 / 60.51 / 83.99 / 68.76), 50M ahead on every subset. Because that set was generated by the same model family as the training data, we also keep a second set built from Qwen3-8B tuples (1,267 items). On it the final systems score 71.76 (5M) and 72.59 (50M), with the 50M system ahead on all four subsets (single→en 79.16 vs 78.71, single→ko 56.92 vs 55.65, mixed→en 84.01 vs 83.37, mixed→ko 65.06 vs 64.06). In round 1 the ordering on this set had been reversed (73.24 vs 71.41): the higher-capacity adapter was fitting the literal Korean style of the earlier references more tightly, and the reversal disappeared once the references were made natural.

5.4 Additional controlled comparisons

50M allocation (same data, 187k rows). On the official validation set the MLP-heavy adapter (attention r=20 / MLP r=60) scores 66.86 against 65.93 for the uniform r=44 adapter, +0.9, almost entirely on Korean targets (single→ko 46.60 → 48.02, mixed→ko 67.54 → 69.35); on our own dev set the two are within 0.2 (74.39 vs 74.54) and on the second dev set the MLP-heavy one is +1.0 (72.37 vs 71.41). **5M allocation (final data).** An MLP-heavy 5M adapter (attention r=2 / MLP r=5, 4,243,456 parameters) scores 67.51 against 67.30 for uniform r=4, equal within noise; we submit the uniform adapter because the 50M system dominates it on more subsets (the MLP-heavy 5M reaches 76.22 on mixed→en, which the 50M system does not). **Regularisation and schedule (50M r=44, own dev set).** Dropout 0.05 + weight decay 0.01: 74.23; a third epoch at lr 8e-5: 74.27; a third epoch for the 5M system: 73.43 (no change); DoRA at r=4 and a learning-rate sweep (1e-4 / 3e-4 / 6e-4) in the pilot: within ±0.3. **Post-processing.** Removing commas from Korean outputs (Section 4) is worth +0.5 on the official set and is applied to all reported numbers. **Checkpoint**

averaging. No gain (5.1). The picture is consistent: at this data scale the adapter configuration changes the score by a few tenths, the training data by several points.

6. Analysis

What the adapters fix. The baseline's low Korean-target scores come mostly from catastrophic outputs rather than weak translation: it echoes or translates the instruction instead of the text, falls into repetition loops, or answers in the wrong language (4–5% of Korean-target items; e.g. for "You have five remaining vacation days this year, ..." it emits "이해를 위해, 이 문장은 다음과 같이 번역됩니다." repeated until the token limit). After fine-tuning these disappear (official validation set: wrong-language outputs 4.0% of Korean-target items at baseline → 0.0% for the 50M system, e.g. "Aunque la máquina no se detuvo, 振動が大きくなったため、작업자가 생산 속도를 낮췄습니다." was answered in Thai by the baseline (2 chrF++) and becomes "기계가 멈추지 않았지만 진동이 커져서 작업자가 생산 속도를 낮췄습니다." (76); length ratio 1.08–1.21 → 1.00–1.03 on every subset; items below 45 chrF++: 218 → 110 of 500; exact matches: 24 → 57; no empty outputs), and the model reliably copies the parts of the input that are already in the target language while translating the rest, e.g. "The blood sampleは届きましたが、検査の依頼書が ありません。" → "혈액 검체는 도착했지만 검사 의뢰서가 없습니다." (exact match; baseline 34) and "The cancellation requestは受け付けましたが、 返金はまだ処理していません。" → "해지 신청은 접수했지만 환불은 아직 처리하지 않았습니다." (83; baseline 16).

Switch-boundary handling. Clause-level alternation and particle-attached insertions are handled well, including three-language inputs: "Could you tell me which 書類を持っていけばよいか, such as my passport, 病院の在職証明書, and two photos?" → "여권, 병원 재직증명서, 사진 두 장이면 어떤 서류를 가져가야 하는지 알려 주시겠습니까?" (87, own dev set). Mixed inputs with three languages score a few points below two-language inputs (own dev set, ko target: 61.7 vs 65.0 chrF++ for the round-1 50M system), and on the official set mixed inputs score higher than single-language inputs for both targets (the mixed items are shorter and partly already in the target language) while Korean targets stay 3–12 points below English targets, i.e. the difficulty is the Korean generation side, not the switch detection.

Remaining failures (items below 45 chrF++: 110 of 500 for the 50M system, 83 of them with Korean target) are dominated by lexical and register choices rather than structural errors: "The solar panelsは稼働していますが、曇りで発電量が落ちています。" → reference "태양광 패널은 가동 중이지만 흐린 날씨로 발전량이 줄었습니다." vs output "태양 패널은 작동하지만 흐리면 발전량이 줄어듭니다." (23) keeps the meaning but shares few character n-grams with the reference. The lowest Korean-target scores on single-language inputs are Spanish sentences in which one noun is mis-sensed: "colocó una señal junto al escalón" → "계단 옆에 신호탄을" ("flare" instead of "sign", 13), "capacidad menor" → "수용 인원이 적습니다" ("fewer people" instead of "smaller load capacity", 14); the backbone's Spanish→Korean lexical knowledge is the bottleneck there. True meaning changes around a language switch concentrate in English-matrix sentences with a Spanish noun phrase inside a negated conditional: "Do not interpret la ausencia de respuestas as agreement unless participants were explicitly given that option." → "응답이 없는 것은 참가자가 그 옵션을 명시적으로 선택한 경우에만 동의를 의미하지 마세요." (24) garbles the scope of "unless", and on our own dev set in Japanese-heavy inputs with English nouns inside Japanese grammar ("Will the reading responseのdeadlineは、Tuesday sectionではなく Thursday section に出席するstudentsでも同じですか" → "화요일 쪽으로 Thursday 쪽에"). Omissions at the switch point are rare: the length ratio is 0.99–1.03 on every subset, and because chrF++ is recall-weighted, staying literal and complete is the right behaviour for this metric.

Budget comparison. The 50M system beats the 5M system on all four subsets of both own dev sets at every data size we tried (5.2, 5.3), with the largest margin on Korean targets (+2.3 mixed→ko, +1.8 single→ko on the official set), consistent with the view that r=4 is capacity-limited for the Korean generation side. Its one shortfall, −1.0 on official single→en, is a sentence-level tie (63 wins / 61 losses of 150) driven by a few Spanish sentences that the larger adapter paraphrases ("The ad received many views" for "generó muchas visitas", reference "generated many visits"); the larger adapter has learned more of the natural English style of the human pairs, which a literal single reference does not reward. The 50M system is also safer (no wrong-language outputs vs 1 for 5M) and more concise on Korean targets, where the metric gains are largest. A learning-rate sweep (1e-4 / 3e-4 / 6e-4) and DoRA at r=4 changed the pilot score by less than ±0.3, so the final systems use plain LoRA (5M: lr 3e-4, 50M: lr 2e-4, cosine, 3% warm-up, effective batch 16, 2 epochs).